

measure theory extra

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August 2026

1 Cantor Set

Let $C_0 = I$ and define $C_{n+1} = \frac{1}{3}(C_n \cup (2 + C_n))$. If $C_n \leq I$, then $C_{n+1} \leq \frac{1}{3}[0, 3] = I$. Further, if $C_{n+1} \leq C_n$, then

$$C_{n+2} = \frac{1}{3}(C_{n+1} \cup (2 + C_{n+1})) \leq \frac{1}{3}(C_n \cup (2 + C_n)) = C_{n+1},$$

so the C_n 's are decreasing. Since $x \mapsto x/3$ is a homeomorphism of $\mathbb{R}$, if C_n is closed, so is C_{n+1} . Thus, $C = \bigcap_{n \geq 0} C_n$ is closed and contained in I .

I claim that C is uncountable. Let $F : \{0, 1\}^{\mathbb{Z}^+} \rightarrow C$ be defined by $F(f) = 2 \sum_{n=1}^{\infty} f(n)/3^n$. It suffices to show that F is injective. First we show that F is well-defined. $F(f) \in C_0$ since $0 \leq F(f) \leq 2 \sum_{n=1}^{\infty} \frac{1}{3^n} = 1$. Assume $F(f) \in C_k$. Then $F(f) = \frac{2}{3}f(1) + \frac{2}{3} \sum_{k=1}^{\infty} \frac{f(k+1)}{3^k} \in \frac{1}{3}(2f(1) + C_k) \leq C_{k+1}$. Thus, $F(f) \in C$. Now we show that F is injective. Suppose $F(f) = F(g)$ with $f \neq g$. Let $a \in \mathbb{Z}^+$ be minimal such that $f(a) \neq g(a)$. WLOG $f(a) = 0$ and $g(a) = 1$. We estimate

$$0 \leq 2 \sum_{n=a+1}^{\infty} \frac{f(n)}{3^n} \leq \frac{1}{3^a}$$

(and have a similar formula for g). Let $x = 2 \sum_{n=1}^{a-1} \frac{f(n)}{3^n}$. Then $F(f) \in [x, x + \frac{1}{3^a}]$, while $F(g) \in [x + \frac{2}{3^a}, x + \frac{3}{3^a}]$. However, the intervals are disjoint, so F is injective.

Let μ^* denote the Lebesgue outer measure. We have $\mu^*(C_{n+1}) \leq \frac{2}{3}\mu^*(C_n)$ by translation invariance and scaling. Since $\mu^*(C_0) = 1$, we have $\mu^*(C_n) \leq (\frac{2}{3})^n$. It follows that $\mu^*(C) \leq \lim_{n \rightarrow \infty} (\frac{2}{3})^n = 0$. This gives the existence of an uncountable set of measure 0.

Let $x \in C$. I claim that there exists a unique $f \in \{0, 1\}^{\mathbb{Z}^+}$ such that $x = F(f)$. It is enough to show existence. First, a lemma.

Lemma: Let $x \in C$. Either $3x \in I$ or $3x \in 2 + I$. If $3x \in I$, then $3x \in C$. If $3x \in 2 + I$, then $3x \in 2 + C$.

Proof: Suppose FTSOC there exist m, n such that $3x \in C_m, 2 + C_n$. Since $C_m, C_n \leq I$, we have that $C_m, 2 + C_n$ are disjoint, absurd. $I = C_0$ and $x \in C_1 \rightarrow 3x \in C_0 \cup (2 + C_0) = I \cup (2 + I)$,

which implies the result.

Now we define f as follows. Let $x_0 = x$. Given x_n , define $x_{n+1} = \{3x_n\} \in C$ and $f(n+1) = \frac{1}{2}[3x_n] \in \{0, 1\}$. I claim that $x = 2 \sum_{k=1}^n \frac{f(k)}{3^k} + \frac{x_n}{3^n}$. The result holds for $n = 0$. Further,

$$\begin{aligned} x &= 2 \sum_{k=1}^n \frac{f(k)}{3^k} + \frac{x_n}{3^n} = 2 \sum_{k=1}^n \frac{f(k)}{3^k} + \frac{2f(n+1) + x_{n+1}}{3^{n+1}} \\ &= 2 \sum_{k=1}^{n+1} \frac{f(k)}{3^k} + \frac{x_{n+1}}{3^{n+1}} \end{aligned}$$

completing the induction. It follows that

$$|x - F(f)| \leq \frac{x_n}{3^n} + \frac{1}{3^n} \leq \frac{2}{3^n}$$

for all n . Taking $n \rightarrow \infty$ gives $F(f) = x$ (convergence is clear). This shows that F is surjective, so $|C| = \mathfrak{c}$.

2 Cantor Function

We define the Cantor function $c : I \rightarrow I$ by

$$c(x) = \begin{cases} \sum_{n=1}^{\infty} \frac{f(n)}{2^n} & \text{where } f = F^{-1}(x) \text{ for } x \in C, \\ \sup_{y \leq x, y \in C} c(y) & \text{for } x \notin C. \end{cases}$$

$0 \in C$, so the sup is taken over a nonempty bounded set. It is clear that c stays in its codomain.

Claim: F is differentiable almost everywhere on I with derivative 0.

Proof: Let $x \in I - C$. Since C is closed, pick $V_{2\epsilon}(x)$ not intersecting C . Note that $V_{2\epsilon}(x) \leq (0, 1)$ since $0, 1 \in C$. Let $z \in [x - \epsilon, x + \epsilon]$. I claim that $[0, z] \cap C = [0, x - \epsilon] \cap C$. We have $[0, x - \epsilon] \cap C \leq [0, z] \cap C$. Also,

$$([0, z] - [0, x - \epsilon]) \cap C = (x - \epsilon, z] \cap C \leq (x - \epsilon, x + \epsilon] \cap C = \emptyset.$$

Thus, equality holds and c is constant on $[x - \epsilon, x + \epsilon]$. In particular, c is differentiable with derivative 0 at x . C is Lebesgue measurable of measure 0. Thus, $\mu(I - C) = \mu(I) - \mu(C) = 1$.

Claim: c is nondecreasing.

Proof: This is by adapting the proof that F is surjective. Let $x < y$. $C \cap [0, x]$ is bounded closed, so this set contains its supremum. Similarly, $C \cap [0, y]$ contains its supremum. Assume $x < y$ in C . Let $f = F^{-1}x, g = F^{-1}y$. Let $S = \{n \in \mathbb{Z}^+ : f(n) \neq g(n)\}$. We have $S \neq \emptyset$ since $f \neq g$. Let $n_0 \in S$ be minimal. Suppose FTSOC $f(n_0) = 1$ and $g(n_0) = 0$. We bound the tails

$2 \sum_{n=n_0}^{\infty} \frac{f(n)}{3^n} \in [\frac{2}{3^{n_0}}, \frac{3}{3^{n_0}}]$ and $2 \sum_{n=n_0}^{\infty} \frac{g(n)}{3^n} \in [\frac{0}{3^{n_0}}, \frac{1}{3^{n_0}}]$, which gives $x > y$. In particular, we showed that c is strictly increasing on C .

Claim: c is continuous.

Proof: Every $y \in I$ has a binary representation. Formally, there exists f such that $y = \sum_{n=1}^{\infty} \frac{f(n)}{2^n}$. Let $F(f) = x$, so $c(x) = y$. Thus, c is surjective. The open sets in I are generated by sets of the form $[0, b)$, (a, b) , $(a, 1]$. Since c is surjective nondecreasing, the preimages have the same form. Thus, c is continuous.

Lebesgue measurable but not Borel construction. Source: <https://www.math3ma.com/blog/lebesgue-but-not-borel>. Let $f : [0, 1] \rightarrow [0, 2]$ be given by $f(x) = c(x) + x$. f is a strictly increasing homeomorphism. We have $f(C) \cup f(I - C) = f(I) = [0, 2]$. I claim that $\mu(f(I - C)) = \mu(I - C) = 1^{**}$. Then $\mu(f(C)) = 2 - 1 = 1$. Pick $N \leq f(C)$ such that N is not measurable. $f^{-1}N \leq C$ is measurable (since C has measure 0). Since f is a homeomorphism and N is not Borel, we have that $f^{-1}N$ is also not Borel.

**Need to show.

3 σ -Algebras

Let $\mathcal{C}$ be a π -system on X . Then $\sigma(\mathcal{C}) = \lambda(\mathcal{C})$.

Proof: We have $\lambda(\mathcal{C}) \leq \sigma(\mathcal{C})$. Define $\lambda_1 = \{A \in \lambda(\mathcal{C}) : A \cap B \in \lambda(\mathcal{C}) \text{ for all } B \in \mathcal{C}\}$. I claim that $\lambda_1 = \lambda(\mathcal{C})$. It suffices to show that λ_1 is a λ -system containing $\mathcal{C}$. $\emptyset \in \lambda_1$. Suppose $A \in \mathcal{C}$. Then $A^c \cap B = B - A \cap B$. Write $X - A^c \cap B = (X - B) \cup (A \cap B) \in \lambda(\mathcal{C})$, so $A^c \cap B \in \lambda(\mathcal{C})$. If $\{A_n\}_{n \in \mathbb{Z}^+} \leq \lambda_1$ are disjoint, then $\bigcup_{n \in \mathbb{Z}^+} A_n \cap B \in \lambda_1$. $\mathcal{C} \leq \lambda_1$ since $\mathcal{C}$ is a π -system.

Now define $\lambda_2 = \{A \in \lambda(\mathcal{C}) : A \cap B \in \lambda(\mathcal{C}) \text{ for all } B \in \lambda_1\}$. I claim that λ_2 is a λ -system containing $\mathcal{C}$. The arguments are analogous. $\emptyset \in \lambda_2$. $X - A^c \cap B = (X - B) \cup (A \cap B) \in \lambda(\mathcal{C})$ and λ_2 is closed under disjoint union. λ_2 contains $\mathcal{C}$ since $\lambda_1 = \lambda(\mathcal{C})$. Thus, $\lambda_2 = \lambda(\mathcal{C})$. Thus, $\lambda(\mathcal{C})$ is closed under intersection and is a σ -algebra.

Explicit form of σ -algebra generated by a collection of sets. Let $\mathcal{E} \leq \mathcal{P}(X)$ and assume $\emptyset, X \in \mathcal{E}$. Define $\mathcal{E}_0 = \mathcal{E}$. Suppose $\mathcal{E}_\alpha$ has been defined. Let $\mathcal{E}_{\alpha+1} = \{\bigcup_{n \in \mathbb{Z}^+} A_n \cup \bigcup_{n \in \mathbb{Z}^+} B_n^c : A_n, B_n \in \mathcal{E}_\alpha\}$. Suppose $\alpha > 0$ is a limit ordinal. Define $\mathcal{E}_\alpha = \bigcup_{\beta < \alpha} \mathcal{E}_\beta$.

I claim that $\sigma(\mathcal{E}) = \bigcup_{\alpha < \omega_1} \mathcal{E}_\alpha$. We proceed by induction. $\mathcal{E}_0 = \mathcal{E} \leq \sigma(\mathcal{E})$. Suppose $\mathcal{E}_\alpha \leq \sigma(\mathcal{E})$. Then $\mathcal{E}_{\alpha+1} \leq \sigma(\mathcal{E})$ by definition of a σ -algebra. Suppose $\mathcal{E}_\beta \leq \sigma(\mathcal{E})$ for all $\beta < \alpha$ for some limit ordinal $\alpha > 0$. Then $\bigcup_{\beta < \alpha} \mathcal{E}_\beta \leq \sigma(\mathcal{E})$ by set properties. Thus, $\bigcup_{\alpha < \omega_1} \mathcal{E}_\alpha \leq \sigma(\mathcal{E})$. Note that we can union over ordinals beyond ω_1 , but don't do so.

I claim that $\mathcal{E}_\alpha \leq \mathcal{E}_\beta$ if $\alpha \leq \beta$. We proceed by induction. Write $\beta = \alpha + \gamma$ for $\gamma \geq 0$. We induct on γ . For $\gamma = 0$, the result holds. Suppose the result for γ . Then $\mathcal{E}_{\alpha+(\gamma+1)} = \mathcal{E}_{(\alpha+\gamma)+1} \geq \mathcal{E}_{\alpha+\gamma}$ ($\mathcal{E}_\alpha \leq \mathcal{E}_{\alpha+1}$ since $\emptyset, X \in \mathcal{E}_\alpha$ for all α). Suppose the result for $\gamma' < \gamma$, where $\gamma > 0$ is a limit ordinal. Then $\alpha + \gamma' > 0$ is limit ordinal and $\mathcal{E}_{\alpha+\gamma} = \bigcup_{\beta < \alpha+\gamma} \mathcal{E}_\beta \geq \mathcal{E}_\alpha$, completing the induction.

Now we show that $\bigcup_{\alpha < \omega_1} \mathcal{E}_\alpha$ is a σ -algebra. Let $A \in \bigcup_{\alpha < \omega_1} \mathcal{E}_\alpha$, so $A \in \mathcal{E}_\beta$ for some $\beta < \omega_1$. Then $A^c \in \mathcal{E}_{\beta+1} \leq \bigcup_{\alpha < \omega_1} \mathcal{E}_\alpha$ by construction. Suppose $\{A_n\}_{n < \omega} \leq \bigcup_{\alpha < \omega_1} \mathcal{E}_\alpha$. Pick β_n such that $A_n \in \mathcal{E}_{\beta_n}$. Let $\beta = \sup_{n < \omega} \beta_n < \omega_1$ (since a countable union of countable sets is countable), so $A_n \in \mathcal{E}_\beta$ for all $n < \omega$ by the previous claim. Thus, $\bigcup_{n < \omega} A_n \in \mathcal{E}_{\beta+1} \leq \bigcup_{\alpha < \omega_1} \mathcal{E}_\alpha$.

Suppose $|\mathcal{E}| \leq \mathfrak{c}$. Then I claim that $|\mathcal{E}_\alpha| \leq \mathfrak{c}$ for all $\alpha < \omega_1$. We induct on α . $|\mathcal{E}_0| = |\mathcal{E}| \leq \mathfrak{c}$. Suppose the result for α . Then $|\mathcal{E}_{\alpha+1}| \leq |\mathcal{E}_\alpha|^{\aleph_0} \leq \mathfrak{c}^{\aleph_0} = 2^{\aleph_0^2} = 2^{\aleph_0} = \mathfrak{c}$ (we use that $2\mathfrak{c} = \mathfrak{c}$ as well). Suppose $0 < \alpha \leq \omega_1$ is a limit ordinal. Then $|\mathcal{E}_\alpha| \leq \mathfrak{c} \cdot |\alpha| \leq \mathfrak{c} \cdot \mathfrak{c} = \mathfrak{c}$, since $|\omega_1| \leq \mathfrak{c}$. Since $\mathcal{B}(\mathbb{R}) = \sigma(\mathcal{E})$ for a countable collection $\mathcal{E}$, we have that $|\mathcal{B}(\mathbb{R})| \leq \mathfrak{c}$. We have the inclusion $\mathbb{R} \rightarrow \mathcal{B}(\mathbb{R})$ by $x \mapsto \{x\}$, so $|\mathcal{B}(\mathbb{R})| = \mathfrak{c}$.

Suppose $A \leq X$ is a subset and $\mathcal{E} \leq \mathcal{P}(X)$. Then $\sigma_A(\mathcal{E}|_A) = \sigma_X(\mathcal{E})|_A$.

Proof: $\sigma_X(\mathcal{E})|_A$ is a sigma algebra of A containing $\mathcal{E}|_A$, so $\sigma_X(\mathcal{E})|_A \geq \sigma_A(\mathcal{E}|_A)$. Conversely, define $\mathcal{A} = \{B \leq X : B \cap A \in \sigma_A(\mathcal{E}|_A)\}$. Suppose $B \in \mathcal{A}$. Then $B^c \cap A = A - B \cap A \in \sigma_A(\mathcal{E}|_A)$. Now let $\{B_n\}_{n \in \mathbb{Z}^+}$ denote a collection of sets in $\mathcal{A}$. Let $B = \bigcup_{n \in \mathbb{Z}^+} B_n$. Then $B \cap A = \bigcup_{n \in \mathbb{Z}^+} B_n \cap A \in \sigma_A(\mathcal{E}|_A)$. Finally, $\emptyset \in \mathcal{A}$, so $\mathcal{A}$ is a σ -algebra. $\mathcal{E} \leq \mathcal{A}$, so $\sigma_X(\mathcal{E}) \leq \mathcal{A}$, so $\sigma_X(\mathcal{E})|_A \leq \mathcal{A}|_A \leq \sigma_A(\mathcal{E}|_A)$.

Corollaries: If $A \leq X$ is a subspace, then $\mathcal{B}(A) = \mathcal{B}(X)|_A$. Suppose $f : A \rightarrow B$ is continuous, where $A, B \leq X$ and $B \in \mathcal{B}(X)$. Let $C \leq B$ be Borel in X . Then $f^{-1}C$ is Borel (in X).

Since C is Borel in X , we have $C \cap B = C$ is Borel in B . Since f is continuous, $f^{-1}C$ is Borel in A . Since $\mathcal{B}(A) = \mathcal{B}(X)|_A$, we have that $f^{-1}C$ is Borel in X .

4 Measurable Functions

Claim: If $\mathcal{N} = \sigma(\mathcal{E})$, then $f : X \rightarrow Y$ is $(\mathcal{M}, \mathcal{N})$ -measurable iff $f^{-1}(E) \in \mathcal{M}$ for all $E \in \mathcal{E}$.

Proof: We show one direction. Let $\mathcal{N}' = \{N \in \mathcal{N} : f^{-1}(N) \in \mathcal{M}\}$, so $\mathcal{N}'$ is a σ -algebra. By definition, $\mathcal{E} \leq \mathcal{N}'$, so $\mathcal{N} = \sigma(\mathcal{E}) \leq \sigma(\mathcal{N}') = \mathcal{N}'$, so $\mathcal{N}' = \mathcal{N}$, ie f is $(\mathcal{M}, \mathcal{N})$ -measurable.

Corr 2.2: If X, Y are topological spaces, every continuous $f : X \rightarrow Y$ is $(\mathcal{B}_X, \mathcal{B}_Y)$ -measurable.

Proof: Continuous functions pull back open sets to open sets.